# NEXUS ACADEMY - LECTURE NOTES

## Chapter 01: Vector Analysis & Resolution

Course: HSC Physics (PHY-101) | Faculty: Nusrat Jahan (Physics Lecturer)

Location: 965/1 Middle Monipur, Mirpur-2, Dhaka-1216

Key Topics Covered:

1. Definition of Vectors and Scalar Quantities
2. Dot Product and Cross Product with 3D Matrix Determinants
3. Resolution of Vectors along Rectangular Components
4. Relative Velocity and River-Boat Crossing Mechanics
5. Board Exam Standard Numerical Problems with Step-by-Step Solutions

Official Study Material for Enrolled Students Only. Nexus ELMS Compliant.